

Rational Fractal Spline

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Introduction to Fractal Interpolation

- ▶ Fractal interpolation is a modern technique in approximation theory used to fit and analyze scientific data.
- ▶ The graph of these FIF can be used to approximate image component such as the profiles of mountains ranges, the tops of clouds and horizons over forests.
- ▶ This method utilizes an iterated function system (IFS) with rational functions of the form $\frac{p_i(x)}{q_i(x)}$.
- ▶ Here, $p_i(x)$ and $q_i(x)$ are cubic polynomials that include two shape parameters.

Advantages of Rational Cubic Fractal IFS

- ▶ The rational cubic IFS provides additional flexibility over classical rational cubic interpolants.
- ▶ This flexibility arises from the inclusion of scaling factors and shape parameters.
- ▶ Classical rational cubic functions are a special case of the proposed fractal interpolants.

Shape Preservation

- ▶ The fractal interpolation scheme is computationally efficient and adaptable (local, moderately local, or global) based on scaling factors and shape parameters.
- ▶ Sufficient conditions on scaling factors and shape parameters ensure that the interpolation function preserves shape:
 - ▶ Monotonicity
 - ▶ Positivity
 - ▶ Convexity

Given Data and IFS

- ▶ Consider a data set $\{(x_i, f_i), i = 1, 2, \dots, n\}$, where $x_1 < x_2 < \dots < x_n$.

- ▶ Define the IFS

$$I^* = \{\mathbb{R}^2; w_i(x, f) = (L_i(x), F_i(x, f)), i = 1, 2, \dots, n - 1\}.$$

- ▶ Where:

$$L_i(x) = a_i x + b_i$$

- ▶ $L_i(x)$ satisfies the condition

$$L_i(x_1) = x_i, L_i(x_n) = x_i + 1.$$

- ▶ $F_i(x, f) = \alpha_i f + r_i(x)$, where:

$$r_i(x) = \frac{p_i(x)}{q_i(x)}$$

- ▶ $p_i(x)$ and $q_i(x)$ are cubic polynomials, and $q_i(x) \neq 0 \quad \forall x \in [x_1, x_n]$.



$$p_i(x) \equiv P_i(\theta) = U_i(1-\theta)^3 + M_i\theta(1-\theta)^2 + N_i\theta^2(1-\theta) + Z_i\theta^3,$$

$$q_i(x) \equiv Q_i(\theta) = (1-\theta)^3 + v_i\theta(1-\theta)^2 + w_i\theta^2(1-\theta) + \theta^3$$

- ▶ Where U_i , M_i , N_i , and Z_i , for $i = 1, 2, \dots, n-1$, are real parameters as follows:

$$U_i = f_i - \alpha_i f_1,$$

$$Z_i = f_{i+1} - \alpha_i f_n,$$

$$M_i = v_i f_i + h_i d_i - \alpha_i [(x_n - x_1)d_1 + f_1 v_i],$$

$$N_i = w_i f_{i+1} - h_i d_{i+1} + \alpha_i [(x_n - x_1)d_n - f_n w_i].$$

- ▶ $|\alpha_i| < a_i$, for $i = 1, 2, \dots, n-1$.

Derivative Conditions

- ▶ Let $F'_i(x, f) = \frac{\alpha_i f + r'_i(x)}{a_i}$, where $r'_i(x)$ is the derivative of $r_i(x)$ with respect to x .
- ▶ Boundary conditions:

$$F_i(x_1, f_1) = f_i, \quad F_i(x_n, f_n) = f_{i+1}$$

$$F'_i(x_1, f_1) = d_i, \quad F'_i(x_n, f_n) = d_{i+1}$$

- ▶ for $i = 1, 2, \dots, n-1$.
- ▶ $h_i = x_{i+1} - x_i$; $i = 1$ to n
- ▶ d_i ($i = 1, 2, \dots, n$) be the derivative values at the knots.
- ▶ The value of ends points is given by:

$$d_1 = \begin{cases} 0, & \text{if } \Delta_1 = 0, \\ \Delta_1 + \frac{(\Delta_1 - \Delta_2)h_1}{h_1 + h_2}, & \text{otherwise.} \end{cases}$$

- ▶ For d_n :

$$d_n = \begin{cases} 0, & \text{if } \Delta_{n-1} = 0, \\ \Delta_{n-1} + \frac{(\Delta_{n-1} - \Delta_{n-2})h_{n-1}}{h_{n-1} + h_{n-2}}, & \text{otherwise.} \end{cases}$$

- ▶ The value of d_i for $i = 2, 3, \dots, n - 1$ is given by:

$$d_i = \begin{cases} 0, & \text{if } \Delta_{i-1} = 0 \text{ or } \Delta_i = 0, \\ \frac{\Delta_i h_{i-1} + \Delta_{i-1} h_i}{h_i + h_{i-1}}, & \text{otherwise.} \end{cases}$$

- ▶ The variable θ is defined as:

$$\theta = \frac{x - x_1}{x_n - x_1}, \quad x \in [x_1, x_n].$$

- ▶ Then the attractor of IFS is the graph of rational cubic FIF.

Shape Preservation in Fractal Interpolation

Problem:

For an arbitrary selection of scaling factors and shape parameters, the rational cubic fractal interpolation function (FIF) may fail to maintain desired shape properties:

- ▶ **Monotonicity:** The curve may oscillate, even if the data is monotonic.
- ▶ **Positivity:** The interpolation could dip below zero, even if the data is positive.
- ▶ **Convexity (Concavity):** The curvature may not reflect the original data's convex or concave shape.

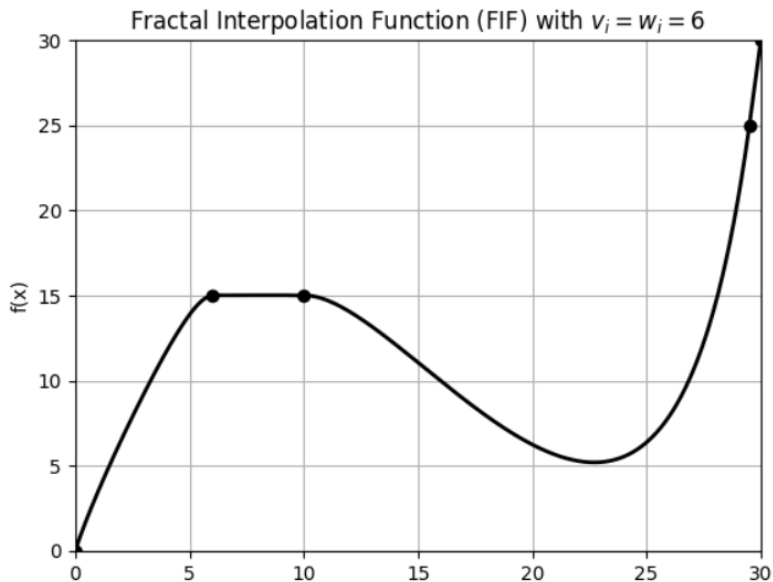
This is very similar to the ordinary spline schemes that do not provide the desired shape features of a data. Thus some mathematical treatment is required to achieve a monotonicity, positivity, and convexity (concavity) preserving rational cubic spline FIF for a given data

Approach to solve the problem

There are two approach to address this:

- ▶ Trail and Error= Manually adjusting the scalling factor and shape parameter but it can work sometime usually not very efficient and precise.
- ▶ Automated Method= Develop automated process to compute the appropriate scalling factor and shape parameter that preserve desired shape properties

Figure



Sufficient conditions for monotonicity

Theorem:

Let $\{(x_i, f_i, d_i), i = 1, 2, \dots, n\}$ be a given monotonic data set. The derivative values must satisfy the necessary conditions for monotonicity:

- ▶ $d_i = d_{i+1} = 0$, for $\Delta_i = 0$;
- ▶ $\text{sgn}(d_i) = \text{sgn}(d_{i+1}) = \text{sgn}(\Delta_i)$ for $\Delta_i \neq 0$.

Now, define the scaling factors α_i for $i = 1, 2, \dots, n - 1$ as:

$$\alpha_i \in \begin{cases} [0, \mu_i] & \text{if } \mu_i < a_i \\ [0, a_i) & \text{if } \mu_i \geq a_i \end{cases}$$

where $\mu_i = \min \left\{ \frac{a_i d_i}{d_1}, \frac{a_i d_{i+1}}{d_n}, \frac{f_{i+1} - f_i}{f_n - f_1} \right\}$.

Additionally, the shape parameters v_i and w_i for $i = 1, 2, \dots, n - 1$ are selected as:

► **Option 1:**

$$v_i = l_i \cdot d_i^* \cdot i^*, \quad w_i = k_i \cdot d_{i+1}^* \cdot i^*, \quad l_i, k_i \in \mathbb{R}^+ \text{ such that } \frac{1}{l_i} + \frac{1}{k_i} \leq 1$$

► **Option 2:**

$$v_i = \eta_i \cdot d_i^* \cdot d_{i+1}^* \cdot i^*, \quad w_i = \nu_i \cdot d_i^* \cdot d_{i+1}^* \cdot i^*$$

where $\eta_i, \nu_i \geq 1$, $d_i^* = d_i - \alpha_i \cdot \frac{d_1}{a_i}$, $d_{i+1}^* = d_{i+1} - \alpha_i \cdot \frac{d_n}{a_i}$, and $i^* = i - \alpha_i \cdot \frac{f_n - f_1}{h_i}$.

Conclusion: From this theorem we find a way to choose scaling factors α_i , shape parameters v_i and w_i , such monotonicity-preserving C^1 -rational cubic FIF, whose graph is the attractor of the rational cubic IFS.

Thank you for your attention!